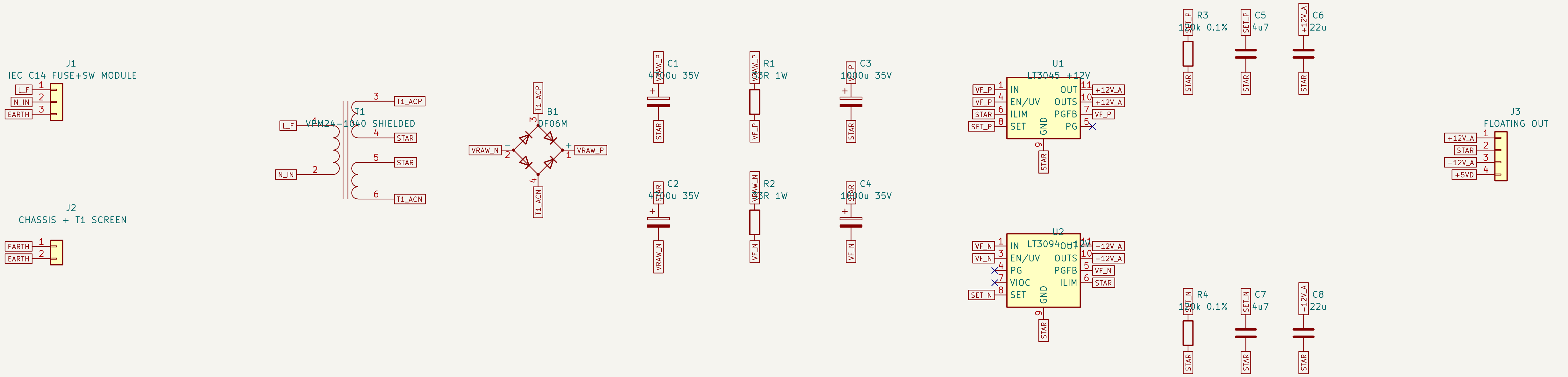
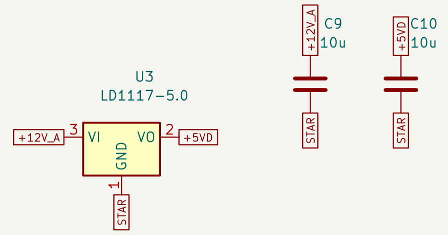
MAINS: Class I. J1 module = C14 inlet + DPST switch + fuse (1A time-lag 5x20). EARTH to chassis stud; T1 screen earthed ONLY at J2 pin 2.

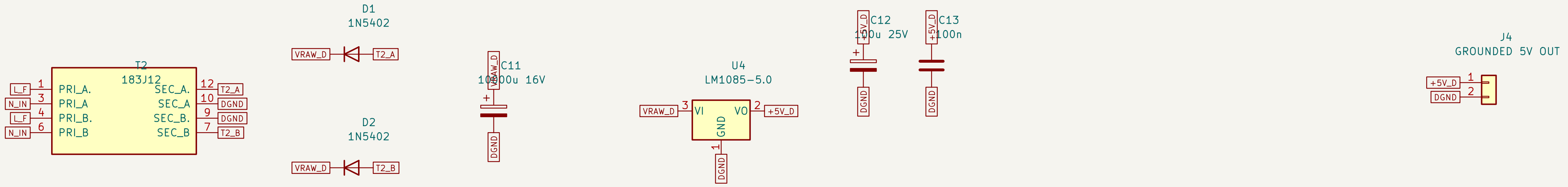
T1 VPM24-1040: dual 115V primaries in PARALLEL (per Triad datasheet dots); dual 12V secondaries in SERIES, junction is floating STAR.



FLOATING RAILS: +/-16V raw -> CRC (33R/1000u) -> LT3045/LT3094 = +/-12.00V at 0.8uVrms. +5VD (LD1117) replaces the dig-board MEE1S0505SC

SMPS - the floating domain is now 100% linear. ECO on dig board: omit PS1/C1/C2, feed +5VD+STAR from JA-4/JA-2.

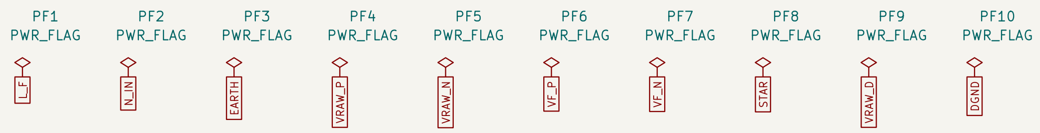




Reg tie-offs closed (per DS + proven Laser_Driver_v3 5V_Laser). VERIFY at bring-up: VPM primary phasing; LM1085 heatsink temp; reg IN pins <19V at 126V line, no load.

Leakage/CM budget: VPM screen diverts primary CM to earth. Target <20pF effective barrier C: route floating umbilical away from mains; twisted+shielded, shield=guard(STAR) at instrument end only.

4700u/10000u Nichicon UPW marked NFND - drop-in alts noted in Description fields.



Grounded 5V via 183J12 + LM1085. Deletes batteries AND dig-board MEE1S0505SC
Floating +/-12V + 5VD via VPM24-1040 medical toroid + LT3045/LT3094

Sheet: /
File: nanovoltmeter_psu.kicad_sch

Title: Nanovoltmeter linear supply — 120V/60Hz, shielded toroid, no bat

Size: A3 | Date: 2026-07-06 | Rev: A
KiCad E.D.A. 10.0.3 | Id: 1/1